

Understanding How Data Centers Impact Surface and Ground Waters | CAES Field Report

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Source Content

The extensive demand for cloud storage, streaming services, and artificial intelligence (AI) in today's digital economy has sparked a massive boom in data center construction. Although concerns and public discussion about this boom primarily focus on the substantial electricity demands of these data centers, they also have a significant local impact on our water resources. These data centers need continuous cooling to prevent their structural components from overheating, which uses an enormous amount of water. In some systems, most of this cooling water evaporates into the atmosphere, while others reuse a large portion of this water, cycling it repeatedly through the cooling systems and eventually discharging it back into the environment. This resource addresses the following key points in a question-and-answer format: - general information about data centers - data centers as an inevitable reality in today's digital economy - potential number of data centers established in Georgia, in the United States, and around the globe - role of water in data center operations - water usage by data centers according to their type and size - how data center operations could alter water availability and affect the quality of our shared surface and underground waters, as well as that of public water systems

Related Content

What Is a Data Center? A data center is a large facility that stores, processes, and distributes digital information using large numbers of computer servers. Modern data centers support cloud computing, streaming, artificial intelligence (AI), social media, banking, healthcare, internet services, and many other online services.

How Many Data Centers Have Been Established in Georgia, in the United States, and Around the Globe? As of May 20, 2026, the Data Center Map website (<https://www.datacentermap.com/usa/>) lists 4,286 data centers nationwide. Georgia ranks fifth in the nation with 213 data centers, after Virginia (603), Texas (461), California (287), and Illinois (228). More data centers are under construction in various U.S. states, including Georgia. Georgia's data centers are heavily concentrated in and around the metro Atlanta area. Major hubs are found in Fulton, Douglas, Newton, Rockdale, and Gwinnett counties (see <https://www.datacenters.com/locations/united-states/georgia>). With approximately 11,400 active data centers worldwide (as of May 20, 2026), the United States has by far the most (38%) of any country globally. The United Kingdom ranks second with 527, followed by Germany with 517, China with 369, and France with 346 (see <https://programs.com/resources/data-center-statistics/>).

How Do Data Centers Use Water? Modern large data centers can contain hundreds of thousands of servers, packed into racks that operate continuously, generating large amounts of heat. As a result, cooling is critical to prevent overheating or even equipment failure. The intensive cooling requirements of data centers use

enormous amounts of electricity and often large volumes of water (U.S. Environmental Protection Agency [EPA], 2026a). Unlike many other industries, data centers do not use water as a raw material; they use it for cooling. The EPA notes that data centers' water demand is increasing rapidly alongside the growth in cloud computing and AI services (U.S. EPA, 2026a). What Are the Different Types of Cooling Systems Used in Data Centers? There are several cooling methods in use by data centers (Hegde, 2026). **Evaporative Cooling (Open-Loop)** A majority of large modern data centers use evaporative cooling because it is energy-efficient, especially in hot climates. The downside is that it directly consumes water (often drawn from a municipal supply). **Closed-Loop Water Cooling** In closed-loop systems, water circulates through sealed pipes or coils to cool servers without being exposed to air. Because the water is not evaporated and is continuously reused (with only small amounts added as needed), these systems can significantly reduce water consumption by up to 70% compared to traditional open, evaporative cooling methods. However, this approach costs more and requires greater system complexity. **Air Cooling (Water-Free)** Many smaller or older data centers rely on air conditioning and chilled-air circulation to remove heat. These are common in cooler climates or where water is scarce, but they require more electricity to run compressors or fans. **Direct Liquid Cooling and Immersion** Direct cooling is an emerging cooling technique that works at the server or chip level. It uses cold liquid (a special coolant) piped directly to server central processing units (CPUs) and graphics processing units (GPUs). Whole servers are submerged in a bath of nonconductive coolant fluid. These methods can significantly reduce water usage. **Free Cooling** In cold climates, some data centers can simply use the outside air or cold water from an adjacent river or sea for cooling. Free cooling is in limited use but is a valuable technique in certain climates or seasons. **Which Cooling System Works Best?** In areas with scarce water, closed-loop water cooling or direct liquid cooling is ideal as those methods use minimal water. In areas with strained power grids, evaporative cooling is favored because it uses less electricity. This highlights a key point: there is a tradeoff between water and energy use. Saving water often means using more electricity for cooling and vice versa. **Is Water Usage the Same or Similar for All Data Centers?** No, water usage in data centers could vary significantly depending on a variety of factors, including: - facility size - cooling technology (evaporative vs non-evaporative coolers) - geographic location (i.e., seasonal temperature fluctuations) - processor advancement (e.g., heat-tolerant GPUs that are in development) **What Type of Water is Used by Data Centers?** Unfortunately, a vast majority of data centers are simply hooked up to the local water mains, meaning most data centers' cooling water comes from potable (drinking-quality) sources like treated municipal water or groundwater. One analysis estimated that around 57% of all data centers use water from potable water supplies (Mytton, 2021). It is also estimated that 80% to 90% of this water is consumed from watersheds—such as lakes, rivers, or aquifers—often the same sources providing drinking water through public water systems (Novva Data Centers, n.d.). Worse yet, most data centers with evaporative cooling systems consume potable water. This removes that water fully out of the local water cycle—usually potable water is not lost entirely to evaporation but is flushed back into municipal water-treatment plants. **What Data Do We Have on Data Center Water Usage?** Globally, 75% to 90% of data centers primarily use water-based cooling (Novva Data Centers, n.d.), meaning only about 10% to 25% use air and/or refrigerants for cooling in water-free systems. A vast majority of large-scale data centers with water-based cooling systems use open-loop evaporative cooling, meaning they consume and evaporate water. This has become standard because it is effective and saves energy. Here are a few facts about data center water usage: - One AI query is estimated to use 16 oz of fresh water (Collier, 2024). ChatGPT alone estimates more than 10 million queries per day. - Browsing or scrolling social media for 1 hr could use 15 oz of fresh water. - A 1-hr video conference call (e.g., Zoom) uses approximately 60 oz of water. - Streaming 1 hr of high-definition video ("1080p"; video that's 1920 by 1080 pixels in size) requires approximately 0.1 to 3 gallons (12 to 384 oz) of water (Travers, 2021). - A medium-sized data center can consume up to 300,000 gallons of water per day for cooling, equivalent to the daily water usage of approximately 1,000 households. - Larger data centers can each use up to 5 million gallons per day, or about 1.8 billion gallons annually, a usage equivalent to a

town of 10,000 to 50,000 people (Yañez-Barnuevo, 2025). - The world's largest data center hub in Loudoun County, VA, used around 1 billion gallons of mostly potable water in 2023, or about 2.75 million gallons per day (Robinson, 2024). - Nationally, all U.S. data centers combined were estimated to consume about 449 million gallons of water per day (1.7 billion liters) in 2021, which is a relatively small amount (0.14%) when compared to the total U.S. water consumption of 322 billion gallons per day (Mytton, 2021). However, data center water use tends to be concentrated in specific regions, so the localized impacts could be substantial even if the national percentage is small. If a data center uses a public water system (the most common water source), its water use could cause domestic water-use restrictions for local residents, especially during a drought. This is why residents often get concerned. Water consumption is expected to continue increasing as data centers grow in number, size, and complexity.

What Water-Related Concerns are Associated with Data Centers?

Aquifer Depletion Many data centers obtain water from municipal groundwater wells or directly from aquifers. Heavy pumping can lower groundwater levels, particularly in drought-prone or heavily developed regions. Potential consequences include: - reduced drinking water availability - lower water tables for private wells - reduced streamflow connected to groundwater

Groundwater Quality Issues Data center cooling systems often use chemical additives to prevent corrosion, scaling, and the growth of algae and bacteria. Common additives may include anti-corrosion/scaling chemicals, biocides, chlorine compounds, glycols, and so forth. Additionally, some nonwater liquid coolants may contain refrigerants and PFAS (also known as “forever chemicals”), requiring special forms of treatment. Fluorinated gases (also known as F-gases, a type of PFAS) may also be used for air cooling. PFAS have negative health impacts and are difficult to remove. If leaks, spills, or improper wastewater disposal occur, contaminants may enter soil and groundwater systems. EPA guidance notes that cooling-system blowdown water can contain chemicals and may require permits or wastewater treatment before discharge (U.S. EPA, 2026b). Blowdown water is the water that's removed from a cooling system to prevent the buildup of minerals, scale, and so on.

Surface Water Withdrawals Some data centers draw water directly from rivers, lakes, or reservoirs. Large withdrawals may: - reduce streamflow during droughts - affect aquatic habitats - increase competition with agricultural and municipal users - alter seasonal water availability These concerns are especially important during drought.

Thermal Pollution of Surface Waters After circulating through cooling systems, water may be discharged back into surface waters at higher temperatures. Warmer discharge water can: - reduce dissolved oxygen levels - stress fish and aquatic organisms - promote algal blooms - cause long-term changes to aquatic ecosystems

Environmental organizations and researchers have warned that thermal discharges from cooling systems could harm ecosystems if not carefully managed (CURE, n.d.).

Chemical Discharge Cooling-tower blowdown water may contain: - salts and minerals - corrosion/scaling inhibitors - biocides - heavy metals from piping systems - refrigerants - PFAS chemicals If inadequately treated, these discharges could affect downstream water quality and aquatic life. In the United States, under the Clean Water Act, facilities that discharge wastewater directly into waterways may require permits under the National Pollutant Discharge Elimination System (NPDES; U.S. EPA, 2026b).

Indirect Impact Data centers also have a large indirect impact on water resources because electricity generation often requires water. Thermoelectric power plants, including fossil fuel and nuclear plants, commonly use water for cooling. Therefore, even air-cooled data centers may indirectly increase water consumption by increasing electricity demand (Siddik et al., 2021). Researchers identified that nearly half of U.S. data center servers are powered partly by plants located in water-stressed regions (Siddik et al., 2021).

What Are Some of the Concerns Communities Have About Data Centers? Common concerns among communities near existing or proposed data centers include: - potential water availability issues or shortages for household use, including drinking water - potential contamination of well water, surface waters, and groundwaters - increased water utility costs - drought resilience - long-term cumulative impacts - transparency in water-use reporting by the data centers

What Can Residents and Communities Do? Test the Water Residents and communities near established or proposed data centers often have concerns about potential contamination

of surface water and/or groundwater (including well water) down the road. Consider testing those waters for the contaminants discussed above to establish a baseline of the local water quality. Future water-test results could be compared with the baseline to evaluate and determine whether water quality is impaired. However, the test results are not going to serve as confirmatory evidence of contamination caused by the data center(s), especially for any litigation. For additional information and water testing services, contact your local county Extension office (<https://extension.uga.edu/county-offices.html>). Reduce Online Activity Residents can help reduce data center water usage by limiting unnecessary online activities (see answers to “What Data Do We Have on Data Center Water Usage?” above). Collaborate With Industry Data centers are an indispensable part of our fast-growing digital economy. However, potential environmental, public health, and social impacts resulting from their operation have not yet been fully assessed. Best management practices for minimizing negative impacts of data centers are still evolving through research. Collaboration between the community and industry is paramount in this area. Are There Public Records Regarding Data Center Water Usage and Related Matters? In general, public records regarding data center water usage are currently limited, though availability is growing. The U.S. EPA monitors aggregate national usage, and some operators release voluntary sustainability reports. However, specific facility-level data is often classified as proprietary or is concealed through local nondisclosure agreements. Georgia Senate Bill 421, (<https://www.legis.ga.gov/legislation/72340>) known as “The Data Center Transparency Act,” is bipartisan legislation introduced in the 2026 legislative session. The bill prevents local governments, authorities, and political subdivisions from using nondisclosure agreements to keep data on the water or electricity usage of private entities (such as data centers) a secret, essentially making this data publicly available. The bill was introduced in late January 2026 and is currently pending in the State and Local Governmental Operations Committee as of May 21, 2026 (see <https://legiscan.com/GA/bill/SB421/2025> for bill updates). References Collier, A. (2024, August 22). Artificial intelligence is using a ton of water. Here’s how to be more resourceful. Water Technologies & Solutions Blog. Retrieved May 21, 2026, from <https://www.watertechnologies.com/blog/artificial-intelligence-using-ton-water-heres-how-be-more-resourceful> CURE. (n.d.). Data centers & water. Retrieved May 21, 2026, from https://curemn.org/wp-content/uploads/2026/02/20251108_Data-Centers-and-Water.pdf Hegde, G. (2026, January 23). Myths vs. reality: Data centers and water usage. Florida Water & Pollution Control Operators Association. Retrieved May 20, 2026, from https://www.fwpcwa.org/content.aspx?page_id=5&club_id=859275&item_id=130961 Mytton, D. (2021). Data centre water consumption. *npj Clean Water*, 4, 11. <https://doi.org/10.1038/s41545-021-00101-w> NOVVA Data Centers. (n.d.). Water-free data center cooling. <https://www.novva.com/data-center-services/water-free-cooling/> Robinson, D. (2024, August 19). Virginia’s datacenters guzzle water like there’s no tomorrow, says FOI-based report. *The Register*. Retrieved May 20, 2026, from <https://www.theregister.com/on-prem/2024/08/19/estimated-datacenter-water-use-shoots-up-in-virginia/596518> Shehabi, A., Smith, S. J., Hubbard, A., Newkirk, A., Lei, N., Siddik, M. A. B., Holecek, B., Koomey, J., Masanet, E., & Sartor, D. (2024, December 19). 2024 United States data center energy usage report. Energy Analysis and Environmental Impacts Division, Lawrence Berkeley National Laboratory. <https://doi.org/10.71468/P1WC7Q> Siddik, M. A. B., Shehabi, A., & Martson, L. (2021). The environmental footprint of data centers in the United States. *Environmental Research Letters*, 16, 064017. <https://doi.org/10.1088/1748-9326/abfba1> Travers, K. (2021, March 5). How can you reduce the environmental impact of your next virtual meeting? MIT Energy Initiative. Retrieved May 21, 2026, from <https://energy.mit.edu/news/how-can-you-reduce-the-environmental-impact-of-your-next-virtual-meeting/> U.S. Environmental Protection Agency. (2026a). Water and data centers. Retrieved May 20, 2026, from <https://www.epa.gov/watersense/water-and-data-centers> U.S. Environmental Protection Agency. (2026b). Water-related permits for data centers. Retrieved May 20, 2026, from <https://www.epa.gov/watersense/water-related-permits-data-centers> Yañez-Barnuevo, M. (2025, June 25). Data centers and water consumption. Environmental and Energy Study Institute. Retrieved May 21, 2026, from <https://www.eesi.org/articles/view/data-centers-and-water-consumption> Young, C. (2026). Clouds on the horizon:

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